Risk Assessment & Mitigation

Severity (a)		Likelihood of Occurrence (b)	
1 – Trivial completion	Slight deviation in line following, no impact on lap	1 – Remote	<10% chance of occurring during the week
2 – Minor	Mouse leaves track but recovers, slight time loss	2 – Unlikely	11-30% chance of happening during the week
3 – Moderate	Unable to set a competitive time	3 – Possible	31-50% chance of happening during the week
4 – Serious	Failure to complete laps consistently	4 – Likely	51-80% chance of happening during the week
5 – Catastrophic	Non-performing mouse or destruction of components.	5 – Very likely	>80% chance of happening during the week

Risk Assessment Matrix

(B)↓ (A)→	Trivial	Minor	Moderate	Serious	Catastrophic
Remote	1	2	3	4	5
Unlikely	2	4	6	8	10
Possible	3	6	9	12	15
Likely	4	8	12	16	20
Very likely	5	10	15	20	25

Risk Rating Bands (A x B)

LOW RISK	MEDIUM RISK	HIGH RISK
Continue, but review periodically to ensure controls remain effective	Continue, but implement additional reasonably practicable controls where possible and monitor regularly	STOP THE ACTIVITY Identify new controls. Activity must not proceed until risks are reduced to a low or medium level

#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)	Additional control/action required	Severity (a)	Likelihood (b)	Risk Rating (a x b)
1	**Integration faults:** Modular circuits do not interface correctly on Friday morning.	Communicate what we are working on and decide what approach we are taking between analogue and digital early on.	4	4	16	Use the "modular construction" approach and establish a clear functional block diagram within the first hour.	4	1	4
2	**Momentum overload:** Mouse gathers too much speed and shoots off the final corner.	Work out the minimum torque required to get over the hill, so there won't be enough momentum to shoot off	5	3	15	Implement real-time position feedback control to slow the "outer" motor or reduce overall PWM duty cycle during curves.	5	1	5
3	**Sensor noise:** 20kHz signal is drowned out by ambient electronic noise, causing loss of track.	Implement the passive resonant bandpass filter and peak detector as specified in the brief.	3	4	12	Conduct extensive tests and keep the bandpass pass wide enough so that it doesn't malfunction when using slightly different signal strengths.	2	3	6
4	**Inadequate drive power:** Mouse lacks torque to ascend the final rising ramp.	Work out the maximum torque the motor can produce considering the weight of our mouse	4	3	12	Test wheel speed under load using the optical tachometer and ensure 6V battery supply is fully charged.	4	1	4
5	**Component failure:** Power MOSFET or Op-amp burnout during testing due to wiring errors.	Use reasonable power supplies.	4	3	12	Double-check wiring with a teammate and use current-limited lab power supplies for initial tests.	4	1	4

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6	**Time management:** Not completing mouse for Friday due to poor communications and teamwork	Set deadlines for tasks to be completed and prioritising between written tasks and technical ones.	5	2	10	Establish clear protocols and team management techniques to allow clear communication of what is being worked on and what needs doing.	5	1	5
7	**Trivial misalignment:** Slight sensor tilt causing a minor 0.2s delay in lap time.	Visual check of the sensors before use	2	2	4	Measure the angle of the sensors prior to use to keep a consistent alignment.	1	1	2